

- 3 Mussels, *Mytilus edulis*, are molluscs that live in sea water on the shorelines of coastal regions around the world. Mussels are a popular seafood and so are widely cultivated. After harvesting the mussels, the mussel farmer must replace them with young mussels. This is often done by collecting young mussels from wild marine ecosystems.

Fig. 3.1 shows mature mussels after harvesting.



**Fig. 3.1**

The slipper limpet, *Crepidula fornicata*, is an invasive alien species introduced into Europe from North America. Slipper limpets compete with the mussels and reduce the yield of mussels. Sometimes slipper limpets are accidentally collected with young mussels. Mussel farmers want to prevent the introduction of slipper limpets when re-stocking their mussel farms.

Fig. 3.2 shows slipper limpets.



**Fig. 3.2**

Scientists investigated the best way to kill slipper limpets without harming the mussels. The scientists put 30 slipper limpets (length > 30 mm) into each of four trays. Each tray of slipper limpets was exposed to a different test condition for a period of three days at a temperature of 12°C to 13°C. The four test conditions are shown in Table 3.1.

### Table 3.1

| test condition |                    | details of test condition                                                                                                                                |
|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1              | air exposure       | leave organisms exposed to the air                                                                                                                       |
| 2              | brine rinse        | at the start of day 1, rinse organisms with a saturated salt (sodium chloride) solution for 5 minutes and then leave exposed to the air                  |
| 3              | repeat brine rinse | at the start of day 1, day 2 and day 3, rinse organisms with a saturated salt (sodium chloride) solution for 5 minutes and then leave exposed to the air |
| 4              | brine soak         | at the start of day 1, soak organisms in a saturated salt (sodium chloride) solution for 1 hour and then leave exposed to the air                        |

This procedure was repeated with one tray of 30 slipper limpets (length < 30 mm) and one tray of 30 mussels (length > 30 mm) for each of the four test conditions.

After three days, the scientists counted the number of dead slipper limpets and dead mussels in each tray.

The scientists repeated the whole investigation at a temperature of 4°C to 5°C, using fresh samples of slipper limpets and mussels.

The results of these tests are shown in Fig. 3.3A and Fig. 3.3B.

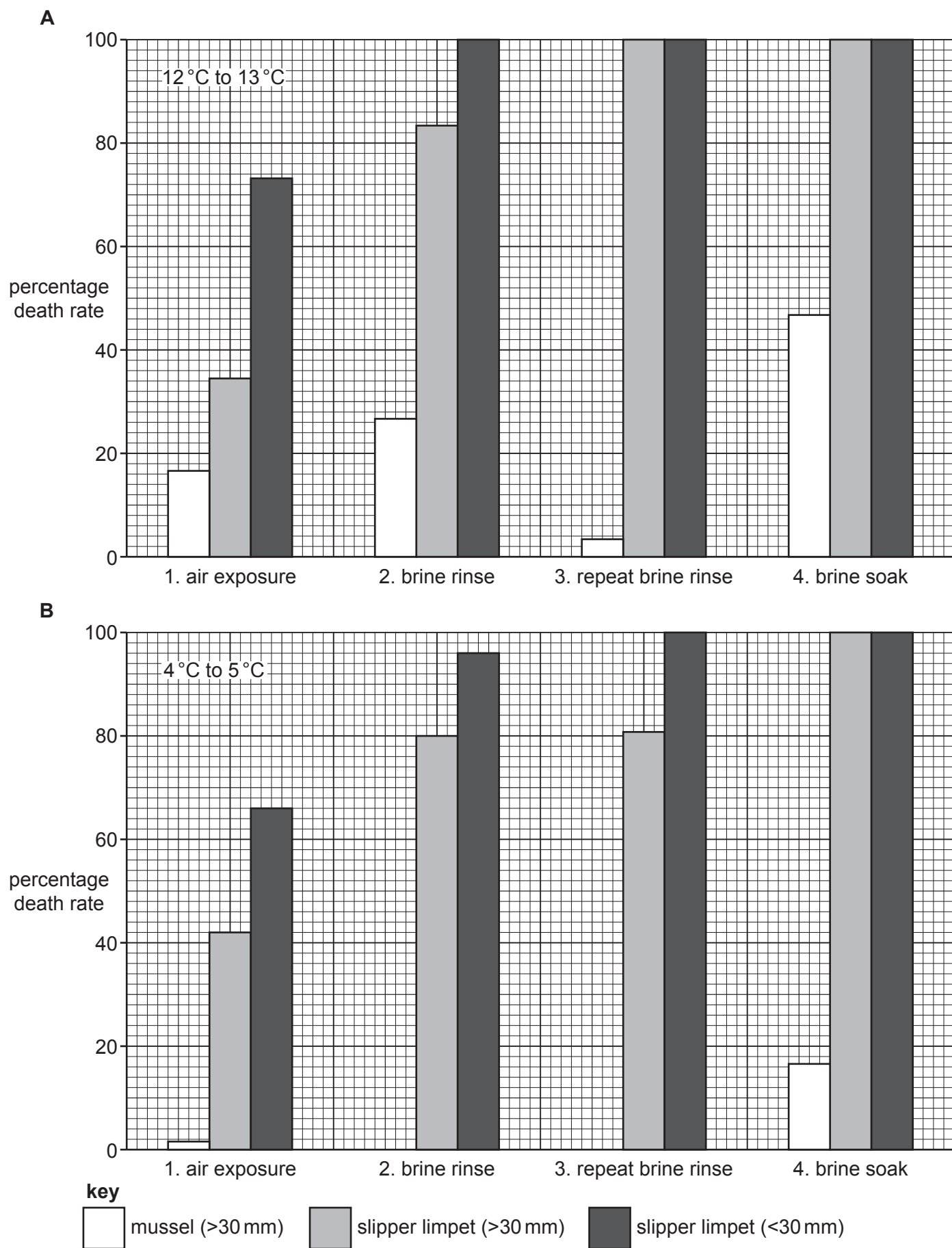
A mussel farmer concluded from the results of the investigation shown in Fig. 3.3 that:

Young mussels used to re-stock the mussel farm should be given a brine soak treatment to kill any slipper limpets that are present.

Evaluate this conclusion.

You should use the data in Fig. 3.3A and Fig. 3.3B to support your answer.

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**Fig. 3.3**